HAMZA SHAHZAD

AI / ML Engineer • Fresh Graduate

Python • PyTorch • TensorFlow • LangChain • Computer Vision • NLP

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer and BS Computer Science (Artificial Intelligence) graduate from The Islamia University of Bahawalpur (CGPA: 2.9/4.0, Jun 2026). Experienced in building end-to-end machine learning and deep learning systems using Python, PyTorch, and TensorFlow. Completed multiple personal and academic projects spanning Computer Vision, NLP, and Generative AI. Skilled in LangChain, RAG pipelines, Hugging Face, and FastAPI. Immediately available for AI/ML roles onsite in Pakistan or remote worldwide.

EDUCATION

BS Computer Science in Artificial Intelligence*Sep 2022 – Jun 2026*

The Islamia University of Bahawalpur, Punjab, Pakistan | CGPA: 2.9 / 4.0 | Status: Graduated

- Key Courses: Machine Learning, Deep Learning, Computer Vision, NLP, Data Structures & Algorithms, Database Systems, Software Engineering

WORK EXPERIENCE

Freelance AI / ML Engineer (Solo)*Jan 2025 – 2026*

Self-Employed • Remote

- Independently designed, built, and delivered AI/ML solutions as a solo developer for personal and client projects.
- Built Computer Vision pipelines using YOLOv8 for real-time object detection and image classification tasks.
- Developed an NLP-based Resume Screening AI tool that automates job-candidate matching using TF-IDF and embeddings.
- Explored and prototyped Generative AI applications using LangChain, RAG pipelines, and open-source LLMs from Hugging Face.
- Deployed AI-powered web tools using Django and FastAPI; experimenting with Docker for containerised deployments.
- Managed complete project lifecycle: problem definition, data collection, preprocessing, modelling, evaluation, and delivery.

PROJECTS

Smart City Issue Detection System Final Year Project (FYP)*2025 – 2026*

Python • YOLOv8 • PyTorch • OpenCV • FastAPI • Django • Docker • Roboflow • GitHub

- Designed and built a full-stack AI Smart City system for automated real-time detection of urban issues Potholes, Garbage, and Broken Streetlights using YOLOv8m (Medium).
- Collected and organised 200–300 images per class across daylight and night conditions; structured raw data into scalable per-class folders (raw_data/pothole, raw_data/garbage, raw_data/streetlight) to support future class expansion.
- Annotated all images on Roboflow with tight bounding boxes; applied data augmentation (horizontal flip, +/-15 degrees rotation, +/-25% brightness) to improve model robustness under real-world lighting and camera variation.
- Trained YOLOv8m on Google Colab T4 GPU for 100 epochs (imgsz=640, batch=16, patience=25); achieved Precision ~0.80, Recall ~0.70, mAP50 ~0.78.
- Engineered modular architecture: backend, frontend, and ml_service modules orchestrated via Docker Compose.

- Served the detection model as a REST API via FastAPI; integrated with Django frontend for live video stream visualisation.
- Built data preprocessing pipeline (prepare_dataset.py) and model evaluation scripts (test_model.py); validated results with confusion matrices and training curves.
- Delivered full project to university supervisor via GitHub: github.com/hamza01055/fyp

Resume Screening AI*2025*

Python • NLP • Scikit-learn • TF-IDF • Django

- Built a full end-to-end NLP pipeline that parses resumes, extracts key skills, and ranks candidates against job descriptions using TF-IDF vectorisation and cosine similarity scoring.
- Wrapped the system in a Django web interface so recruiters can upload resumes and receive ranked shortlists instantly.

Real-Time Object Detection (YOLOv8)*2026*

Python • YOLOv8 • OpenCV • Django

- Implemented a real-time object detection system using YOLOv8 capable of processing live video streams and images.
- Fine-tuned pre-trained YOLOv8 weights on a custom dataset; integrated into a Django web app for browser-based demos.

Generative AI Document Q&A (RAG Pipeline)*2026*

Python • LangChain • FAISS • Hugging Face • FastAPI

- Designed a Retrieval-Augmented Generation (RAG) pipeline answering user queries from custom document collections using FAISS vector store and sentence-transformer embeddings.
- Exposed the Q&A system as a REST API via FastAPI, with plans to containerise using Docker for production readiness.

Image Classification using Deep Learning*2026*

Python • PyTorch • CNN • Transfer Learning (ResNet, VGG)

- Trained CNN models from scratch and using transfer learning (ResNet, VGG) for image classification; performed data augmentation, hyperparameter tuning, and full evaluation with confusion matrices and accuracy/loss curves.

Sentiment Analysis using NLP*2026*

Python • TensorFlow • LSTM • Pandas

- Built an LSTM-based sentiment classification model trained on a labelled product review dataset; achieved over 88% test accuracy with tokenisation, stopword removal, and sequence padding.

TECHNICAL SKILLS

Languages	Python, SQL ,C++,Java
ML / DL Frameworks	PyTorch, TensorFlow, Scikit-learn, Keras, YOLOv8
Data & Analysis	Pandas, NumPy, Matplotlib, Seaborn
Computer Vision	OpenCV, YOLOv8, CNN, Transfer Learning (ResNet, VGG)
NLP & Gen AI	NLTK, spaCy, LangChain, RAG, FAISS, Hugging Face, LLMs
Web & Deployment	FastAPI, Django, REST APIs, Docker, Roboflow
Tools	Git, GitHub, Jupyter, VS Code, Google Colab

AVAILABILITY & PREFERENCES

Immediately available • Open to AI/ML Engineer, Data Scientist, and Python Developer roles • Full-time onsite (Pakistan) or remote worldwide